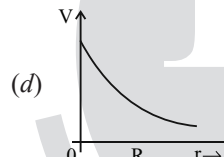
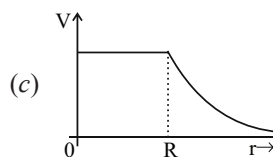
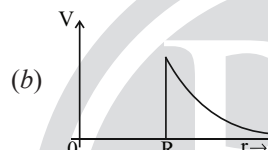
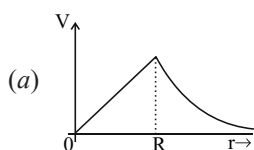


Electrostatic Potential and Capacitance

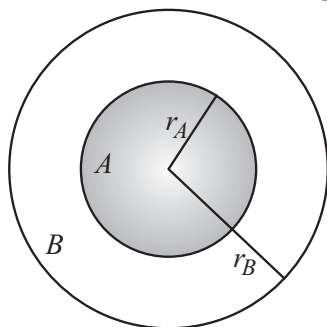
Single Correct Type Question

1. Which of the following graph correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre?
[31 Jan 2023 (Shift-I)]



2. In the figure, the inner (shaded) region A represents a sphere of radius $r_A=1$, within which the electrostatic charge density varies with the radial distance r from the center as $\rho_A = kr$, where k is positive. In the spherical shell B of outer radius r_B , the electrostatic charge density varies as $\rho_B = \frac{2k}{r}$. Assume that dimensions are taken care of. All physical quantities are in their SI units.

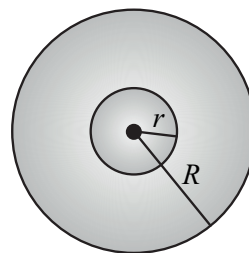
[JEE Adv, 2022]



Which of the following statement(s) is(are) correct?

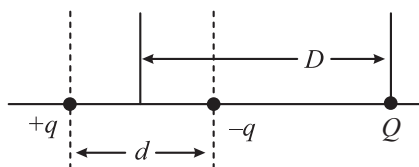
- (a) If $r_B = \sqrt{\frac{3}{2}}$, then the electric field is zero everywhere outside B
- (b) If $r_B = \frac{3}{2}$, then the electric potential just outside B is $\frac{k}{\epsilon_0}$
- (c) If $r_B = 2$, then the total charge of the configuration is $15\pi k$
- (d) If $r_B = \frac{5}{2}$, then the magnitude of the electric field just outside B is $\frac{13\pi k}{\epsilon_0}$
3. A charge Q is distributed over two concentric conducting thin spherical shells radii r and R ($R > r$). If the surface charge densities on the two shells are equal, the electric potential at the common center is

[2 Sep, 2020 (Shift-II)]

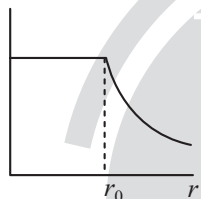
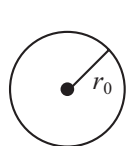


- (a) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)Q}{(R^2+r^2)}$ (b) $\frac{1}{4\pi\epsilon_0} \frac{(R+2r)Q}{2(R^2+r^2)}$
- (c) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)Q}{2(R^2+r^2)}$ (d) $\frac{1}{4\pi\epsilon_0} \frac{(2R+r)Q}{(R^2+r^2)}$

4. A system of three charges are placed as shown in the figure. If $D \gg d$, the potential energy of the system is best given by **[9 April, 2019 (Shift-I)]**



- (a) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right]$ (b) $\frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$
 (c) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} + \frac{2qQd}{D^2} \right]$ (d) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{2D^2} \right]$
5. The given graph shows variation (with distance r from center) of **[11 Jan, 2019 (Shift-I)]**



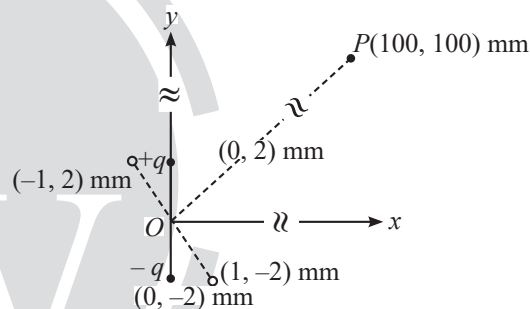
- (a) Potential of a uniformly charged spherical shell
 (b) Electric field of a uniformly charged sphere
 (c) Electric field of uniformly charged spherical shell
 (d) Potential of a uniformly charged sphere
6. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1) / V_2 (on S_2) of the electrostatic potentials on each sphere is **[8 Jan, 2020 (Shift-II)]**
- (a) $\left(\frac{R_1}{R_2}\right)^2$ (b) $\left(\frac{R_1}{R_2}\right)^3$ (c) $\left(\frac{R_2}{R_1}\right)$ (d) $\frac{R_1}{R_2}$
7. Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge $(+q)$ each, while 2, 4, 6, 8, 10 have charge $(-q)$ each. The potential V and the electric field E at the center of the circle are respectively. (Take $V=0$ at infinity) **[5 Sep, 2020 (Shift-II)]**

- (a) $V=0; E=0$ (b) $V=\frac{10q}{4\pi\epsilon_0 R}; E=\frac{10q}{4\pi\epsilon_0 R^2}$
 (c) $V=\frac{10q}{4\pi\epsilon_0 R}; E=0$ (d) $V=0; E=\frac{10q}{4\pi\epsilon_0 R^2}$

8. The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are $A = 20$ SI unit and $B = 10$ SI unit. If the potential at $x = 1$ is V_1 and that at $x = -5$ is V_2 , then $V_1 - V_2$ is:

[8 April, 2019 (Shift-II)]

- (a) $180 V$ (b) $-520 V$ (c) $320 V$ (d) $-48 V$
9. An electric dipole is formed by two charges $+q$ and $-q$ located in xy -plane at $(0, 2)$ mm and $(0, -2)$ mm, respectively, as shown in the figure. The electric potential at point $P(100, 100)$ mm due to the dipole is V_0 . The charges $+q$ and $-q$ are then moved to the points $(-1, 2)$ mm and $(1, -2)$ mm, respectively. What is the value of electric potential at P due to the new dipole? **[JEE Adv, 2023]**



- (a) $V_0/4$ (b) $V_0/2$
 (c) $V_0/\sqrt{2}$ (d) $3V_0/4$
10. A point dipole $\vec{p} = -p_0\hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively (Take $V=0$ at infinity) **[12 April, 2019 (Shift-I)]**

- (a) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$ (b) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$
 (c) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$ (d) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

11. A parallel plate capacitor has plate area 40 cm^2 and plate separation 2 mm . The space between the plates is filled with a dielectric medium of thickness 1 mm and dielectric constant 5 . The new capacitance of the system is : **[25 Jan, 2023 (Shift-I)]**

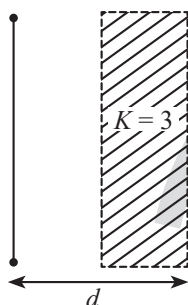
- (a) $24\epsilon_0 F$ (b) $\frac{3}{10}\epsilon_0 F$ (c) $\frac{10}{3}\epsilon_0 F$ (d) $10\epsilon_0 F$

12. Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is: [25 July, 2022 (Shift-II)]

- (a) $\frac{n}{n-1}$ (b) $\frac{2n}{2n+1}$
(c) $\frac{n+1}{n}$ (d) $\frac{2n+1}{2n}$

13. A parallel plate capacitor with plate area A and plate separation $d = 2\text{m}$ has a capacitance of $4\mu\text{F}$. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant $K = 3$ (as shown in figure) will be:

[26 June, 2022 (Shift-II)]

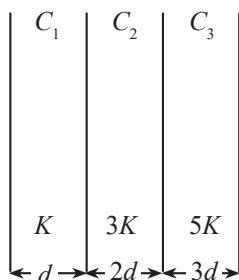


- (a) $2\mu\text{F}$ (b) $32\mu\text{F}$
(c) $6\mu\text{F}$ (d) $8\mu\text{F}$

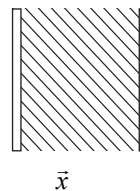
14. In the reported figure, a capacitor is formed by placing a compound dielectric between the plates of parallel plate capacitor. The expression for the capacity of the said capacitor will be: (Given area of plate = A)

[27 July, 2021 (Shift-I)]

- (a) $\frac{15 K \epsilon_0 A}{34 d}$
(b) $\frac{15 K \epsilon_0 A}{6 d}$
(c) $\frac{25 K \epsilon_0 A}{6 d}$
(d) $\frac{9 K \epsilon_0 A}{6 d}$



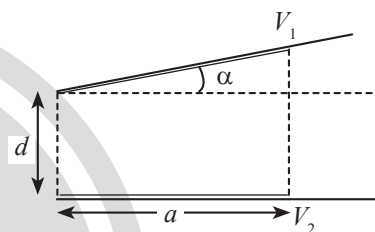
15. A parallel plate capacitor has plates of area A separated by distance ' d ' between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = k(1 + \alpha x)$ where ' x ' is the distance measured from one of the plates. IF $(\alpha d) \ll 1$, the total capacitance of the system is best given by the expression: [7 Jan, 2020 (Shift-I)]



- (a) $\frac{AK\epsilon_0}{d} \left(1 + \frac{\alpha d}{2}\right)$ (b) $\frac{A\epsilon_0 K}{d} \left(1 + \frac{\alpha^2 d^2}{2}\right)$
(c) $\frac{AK\epsilon_0}{d} (1 + \alpha d)$ (d) $\frac{A\epsilon_0 K}{d} \left[1 + \left(\frac{\alpha d}{2}\right)^2\right]$

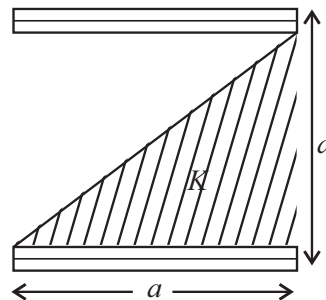
16. A capacitor is made of two square plates each of side ' a ' making a very small angle α between them, as shown in figure. The capacitance will be close to:

[8 Jan, 2020 (Shift-II)]



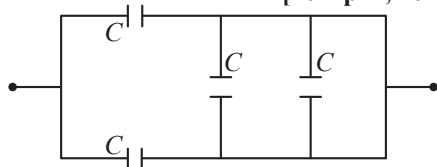
- (a) $\frac{\epsilon_0 a^2}{d} \left(1 + \frac{\alpha a}{d}\right)$ (b) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{4d}\right)$
(c) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{3\alpha a}{2d}\right)$ (d) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$

17. A parallel plate capacitor is made of two square plates of side ' a ', separated by a distance d ($d \ll a$). The lower triangular portion is filled with a dielectric of dielectric constant K , as shown in the figure. Capacitance of this capacitor is [9 Jan, 2019 (Shift-I)]

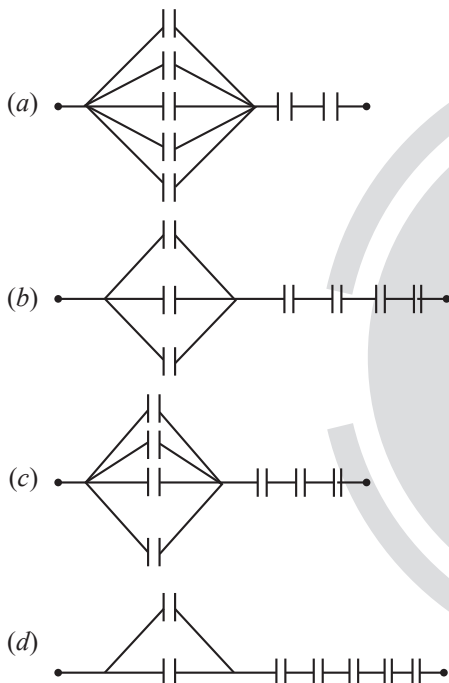


- (a) $\frac{K\epsilon_0 a^2}{d(K-1)} \ln K$ (b) $\frac{K\epsilon_0 a^2}{d} \ln K$
(c) $\frac{K\epsilon_0 a^2}{2d(K+1)}$ (d) $\frac{1}{2} \frac{K\epsilon_0 a^2}{d}$

18. The equivalent capacitance of the combination shown is
[10 April, 2023 (Shift-I)]



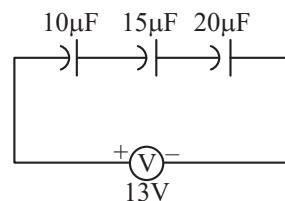
- (a) $\frac{C}{2}$ (b) $4C$ (c) $2C$ (d) $\frac{5}{3}C$
19. Seven capacitors, each of the capacitance $2\mu F$, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right)\mu F$. Which of the combinations, shown in figures below, will achieve the desired value?
[11 Jan, 2019 (Shift-II)]



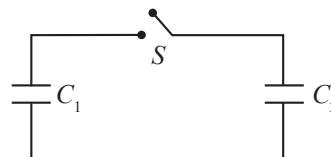
20. Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is:

[5 Sep, 2020 (Shift-I)]

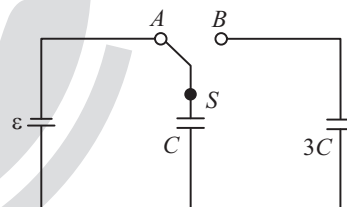
- (a) $\frac{3}{2}CV^2$ (b) $\frac{9}{2}CV^2$
(c) $\frac{25}{6}CV^2$ (d) zero
21. The charge on capacitor of capacitance $15\mu F$ in the figure given below is:
[26 June, 2022 (Shift-II)]



22. Two capacitors having C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S . The amount of charge on the capacitor C_2 , after equilibrium, is:
[26 June, 2022 (Shift-I)]



- (a) $\frac{C_1 C_2}{(C_1 + C_2)} V$ (b) $\frac{(C_1 + C_2)}{C_1 C_2} V$
(c) $(C_1 + C_2) V$ (d) $(C_1 - C_2) V$
23. In the figure shown, after the switch 'S' is turned from position 'A' to position 'B', the energy dissipated in the circuit in terms of capacitance ' C ' and total charge ' Q ' is
[12 Jan, 2019 (Shift-I)]

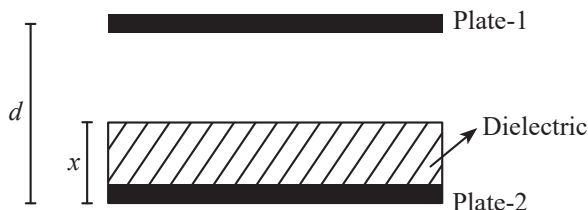


- (a) $\frac{3}{8} \frac{Q^2}{C}$
(b) $\frac{1}{8} \frac{Q^2}{C}$
(c) $\frac{5}{8} \frac{Q^2}{C}$
(d) $\frac{3}{4} \frac{Q^2}{C}$
24. A parallel plate capacitor has $1\mu F$ capacitance. One of its two plates is given $+2\mu C$ charge and the other plate, $+4\mu C$ charge. The potential difference developed across the capacitor is:
[8 April, 2019 (Shift-II)]

- (a) 3V (b) 1V
(c) 5V (d) 2V

Integer Type Question

25. A parallel plate capacitor with plate area A and plate separation d is filled with a dielectric material of dielectric constant $K = 4$. The thickness of the dielectric material is x , where $x < d$.

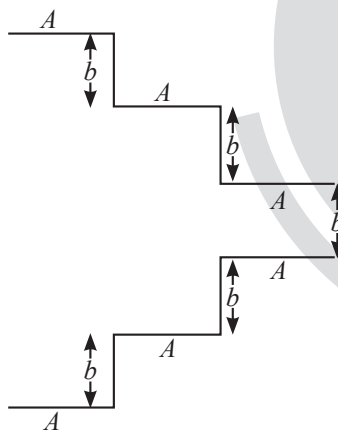


Let C_1 and C_2 be the capacitance of the system for $x = \frac{1}{3}d$ and $x = \frac{2d}{3}$, respectively. If $C_1 = 2\mu F$ the value of C_2 is _____ μF

[6 April, 2023 (Shift-I)]

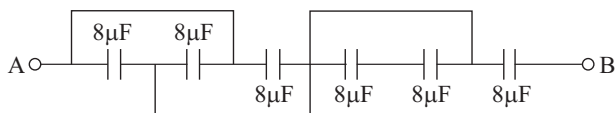
26. A parallel plate capacitor is made up of stair like structure with a plate area A of each stair and that is connected with a wire of length b , as shown in the figure. The capacitance of the arrangement is $\frac{x}{15} \frac{\epsilon_0 A}{b}$, the value of x is _____

[27 June, 2022 (Shift-II)]



27. The equivalent capacitance between points A and B in below shown figure will be _____ μF .

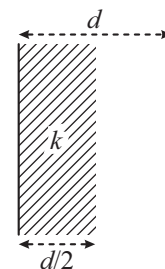
[25 June, 2022 (Shift-I)]



28. In a parallel plate capacitor set up, plate area of capacitor is 2 m^2 and the plates are separated by 1 m . If the space

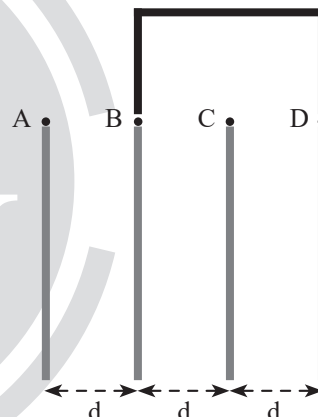
between the plates are filled with a dielectric material of thickness 0.5 m and area 2 m^2 (See fig) the capacitance of the set-up will be _____ ϵ_0 . (Dielectric constant of the material = 3.2) (Round off to the Nearest Integer)

[16 March, 2021 (Shift-II)]



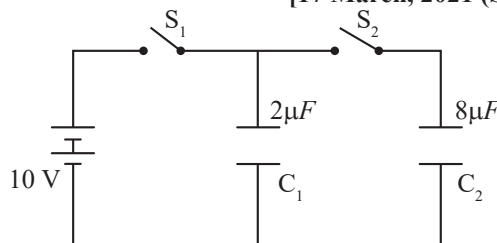
29. Four identical rectangular plates with length, $\ell = 2$ and breadth, $b = \frac{3}{2} \text{ cm}$ are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\epsilon_0}{d}$. The value of x is _____ (Round off to the nearest integer)

[17 March, 2021 (Shift-I)]



30. A $2\mu F$ capacitor C_1 is first charged to a potential difference of 10 V using a battery. Then the battery is removed and the capacitor is connected to an uncharged capacitor C_2 of $8\mu F$. The charge in C_2 on equilibrium condition is _____ μC . (Round off to the Nearest integer)

[17 March, 2021 (Shift-II)]



ANSWER KEY

1. (c)	2. (b)	3. (a)	4. (a)	5. (a)	6. (a)	7. (a)	8. (a)	9. (b)	10. (c)
11. (c)	12. (a)	13. (c)	14. (a)	15. (a)	16. (d)	17. (a)	18. (c)	19. (b)	20. (a)
21. (a)	22. (a)	23. (a)	24. (b)	25. [3]	26. [23]	27. [6]	28. [3]	29. [2]	30. [16]

EXPLANATIONS

1. (c) Conceptual

2. (b) For $r \in [0, 1]$; $\rho = kr$

$$\text{For } r \in [1, r_B]; \rho = \frac{2k}{r}$$

Total charge of the configuration is q

$$q = \int_0^1 (kr) 4\pi r^2 dr + \int_1^{r_B} \frac{2k}{r} 4\pi r^2 dr$$

$$q = k\pi + 4k\pi(r_B^2 - 1)$$

$$\text{For } r_B = \sqrt{\frac{3}{2}}; q = k\pi + 2k\pi = 3k\pi$$

Electric field just outside B is

$$E = \frac{1}{4\pi\epsilon_0} \frac{(3\pi k)}{r_B^2} = \frac{k}{2\epsilon_0}$$

$$\text{For } r_B = \frac{3}{2}; q = k\pi + 2k\pi = 3k\pi$$

Electric field just outside B is

$$E = \frac{1}{4\pi\epsilon_0} \frac{(3\pi k)}{r_B^2} = \frac{k}{2\epsilon_0}$$

$$\text{For } r_B = \frac{3}{2}; q = k\pi + 5k\pi = 6k\pi$$

$$\text{Potential just outside B is, } V = \frac{(6\pi k)}{4\pi\epsilon_0 \left(\frac{3}{2}\right)} = \frac{k}{\epsilon_0}$$

$$\text{For } r_B = 2; q = k\pi + 12k\pi = 13k\pi$$

$$\text{For } r_B = \frac{5}{2}; q = k\pi + 21k\pi = 22k\pi$$

Magnitude of E just outside B is

$$\frac{1}{4\pi\epsilon_0} \frac{(22\pi k)}{\left(\frac{25}{4}\right)} = \frac{22k}{25\epsilon_0}$$

3. (a) Let charges on shells be q_1 and q_2

$$q_1 + q_2 = Q \quad \dots(i)$$

$$\frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2} \quad \dots(ii)$$

$$\text{We get } q_1 = \frac{r^2}{r^2 + R^2} Q, q_2 = \frac{R^2}{r^2 + R^2} Q$$

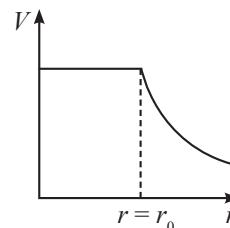
$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{R} \right) = \frac{1}{4\pi\epsilon_0} \frac{(R+r)}{(R^2 + r^2)} Q$$

$$4. (a) U = -\frac{Kq^2}{d} + QV$$

$$V = -\frac{KP}{D^2} = -\frac{Kqd}{D^2}; U = -\frac{Kq^2}{d} - \frac{KQqd}{D^2}$$

5. (a) For spherical shell $V = \frac{K \cdot q}{r_0}, r \leq r_0$

$$= \frac{Kq}{r}, r > r_0$$



$$6. (a) \frac{E_1}{E_2} = \frac{R_1}{R_2}$$

$$\frac{V_1}{V_2} = \frac{E_1 R_1}{E_2 R_2} = \frac{R_1}{R_2} \times \frac{R_1}{R_2} = \left(\frac{R_1}{R_2} \right)^2$$

7. (a) $\Sigma q = 0$

$\Rightarrow V = 0$

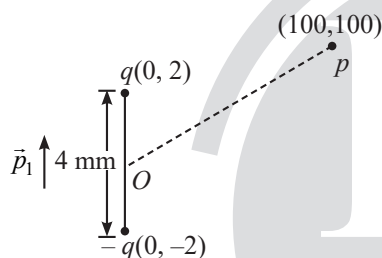
Also, field of one charge will get cancelled due to another symmetrical charge in front of it.

8. (a) $dV = -\vec{E} \cdot d\vec{r} = -(Ax + B)dx$

$$\int_{V_2}^{V_1} dV = \int_{-5}^1 -(Ax + B)dx$$

$$\begin{aligned} V_1 - V_2 &= \left(-A \frac{x^2}{2} - Bx \right)_{-5}^1 \\ &= \left(-\frac{A}{2} - B \right) + \left(\frac{A}{2} 25 + B(-5) \right) \\ &= 12A - 6B = 240 - 60 = 180V \end{aligned}$$

9. (b) Case 1: Initial position of dipole $\{q(0, 2) \text{ and } -q(0, -2)\}$



$$\vec{p}_1 = 4q \hat{j}$$

$$\vec{OP} = \vec{r} = 100\hat{i} + 100\hat{j}$$

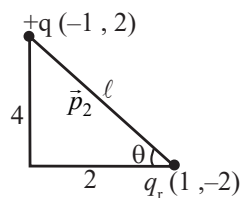
$$V_P = \frac{k\vec{p}_1 \cdot \vec{r}}{|\vec{r}|^3}$$

Potential at P due to dipole

$$= \frac{k\{(4q\hat{j}) \cdot (100\hat{i} + 100\hat{j})\}}{(100\sqrt{2})^3}$$

$$V_{P1} = \frac{400kq}{(100\sqrt{2})^3} \quad \dots(i)$$

Case 2 : Final position of diode $\{q(-1, 2) \text{ and } -q(1, -2)\}$



$$\tan \theta = 3 \sin \theta = \frac{4}{5}, \cos \theta = \frac{2}{5}$$

$$\vec{p}_2 = 5q(-\cos \theta \hat{i} + \sin \theta \hat{j})$$

$$\vec{r} = 100\hat{i} + 100\hat{j}$$

Potential at P due to dipole

$$\begin{aligned} V_{P2} &= \frac{K\vec{p}_2 \cdot \vec{r}}{|\vec{r}|^3} \\ &= \frac{K\{5q(-\cos \theta \hat{i} + \sin \theta \hat{j}) \cdot (100\hat{i} + 100\hat{j})\}}{(100\sqrt{2})^3} \end{aligned}$$

$$V_{P2} = \frac{500Kq(-\cos \theta + \sin \theta)}{(100\sqrt{2})^3} \quad \dots(ii)$$

Using equation (i) and (ii)

$$\frac{V_{P2}}{V_{P1}} = \frac{500Kq(-\cos \theta + \sin \theta)}{(400Kq)} = \frac{5}{4} \left(\frac{-2}{4} + \frac{4}{5} \right)$$

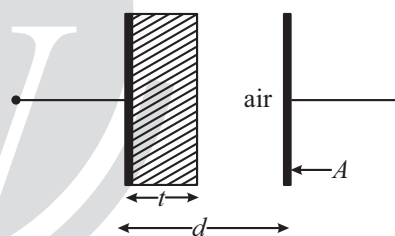
$$V_{P2} = \frac{V_{P1}}{2} = \frac{V_0}{2}$$

10. (c) $\vec{E} = K \frac{\vec{p}}{r^3} \sqrt{3\cos^2 \theta + 1}$

$$\Rightarrow \theta = \pi/2 \text{ (0, d, 0)}$$

$$\therefore \vec{E} = \frac{-k\vec{p}}{d^3}$$

11. (c)



Above arrangement is similar as two capacitors in series combination so

$$\Rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

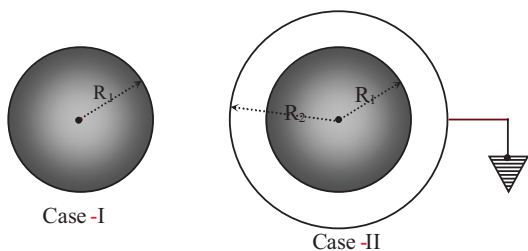
$$= \frac{1}{\frac{K \epsilon_0 A}{t}} + \frac{1}{\frac{\epsilon_0 A}{d-t}}$$

$$= \frac{1 \times 10^{-3}}{5 \epsilon_0 \times 40 \times 10^{-4}} + \frac{1 \times 10^{-3}}{\epsilon_0 \times 40 \times 10^{-4}}$$

$$\frac{1}{C_{eq}} = \frac{1}{20 \epsilon_0} + \frac{1}{4 \epsilon_0}$$

$$C_{eq} = \frac{20 \times 4 \epsilon_0}{24} = \frac{10 \epsilon_0}{3} F$$

12. (a)



Initially, $C_1 = 4\pi\epsilon_0 R_1$ and

$$\text{Finally, } C_2 = \frac{4\pi\epsilon_0 R_2 R_1}{R_2 - R_1}$$

According to question, $C_2 = nC_1 \Rightarrow \frac{C_2}{C_1} = n$

$$\therefore \frac{4\pi\epsilon_0 R_1 R_2}{R_2 - R_1} \times \frac{1}{4\pi\epsilon_0 R_1} = n$$

$$\frac{R_2}{R_2 - R_1} = n$$

$$nR_2 - nR_1 = R_2$$

$$(n-1)R_2 = nR_1$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{n}{n-1}$$

13. (c) Given, $C = \frac{\epsilon_0 A}{d} = 4\mu F$

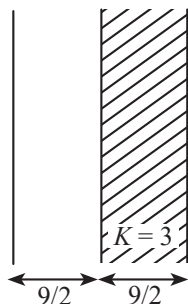
Formula of capacitance, $C = \frac{\epsilon_0 A}{d}$

$$\therefore C_1 = \frac{\epsilon_0 A}{\frac{d}{2}} = 4 \times 2 = 8\mu F$$

$$C_2 = \frac{k\epsilon_0 A}{d/2} = 3 \times 2 \times 4 = 24\mu F$$

Since, C_1 and C_2 are in series, therefore.

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{8 \times 24}{8 + 24} = \frac{8 \times 24}{32} = 6\mu F$$



14. (a) Capacitors, C_1 , C_2 and C_3 are in series therefore,

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{eq}} = \frac{1}{\left(\frac{kA\epsilon_0}{d}\right)} + \frac{1}{\left(\frac{3kA\epsilon_0}{2d}\right)} + \frac{1}{\left(\frac{5kA\epsilon_0}{3d}\right)}$$

$$\frac{1}{C_{eq}} = \frac{d}{kA\epsilon_0} + \frac{2d}{3kA\epsilon_0} + \frac{3d}{5kA\epsilon_0}$$

$$\frac{1}{C_{eq}} = \frac{(15+10+9)d}{15kA\epsilon_0}$$

$$C_{eq} = \frac{15kA\epsilon_0}{34d}$$

15. (a) Capacitance of element, $dc = \frac{K(1+\alpha x)\epsilon_0 A}{dx}$

As all elements are in series, so equivalent capacitance

$$\sum \frac{1}{c} = \int \frac{1}{dc} = \int_0^d \frac{dx}{k\epsilon_0 A(1+\alpha x)}$$

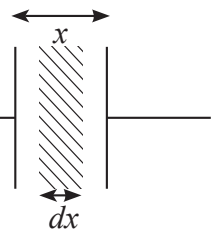
$$\frac{1}{C} = \frac{1}{K\epsilon_0 A\alpha} \ln\left(\frac{1+\alpha d}{1}\right)$$

Given: $\alpha d \ll 1$

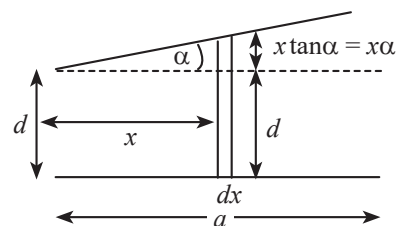
$$\frac{1}{C} = \frac{1}{K\epsilon_0 A\alpha} \left(\alpha d - \frac{\alpha^2 d^2}{2}\right);$$

$$\Rightarrow \frac{1}{C} = \frac{d}{K\epsilon_0 A} \left(1 - \frac{\alpha d}{2}\right)$$

$$\Rightarrow C = \frac{AK\epsilon_0}{d} \left(1 + \frac{\alpha d}{2}\right)$$



16. (d)



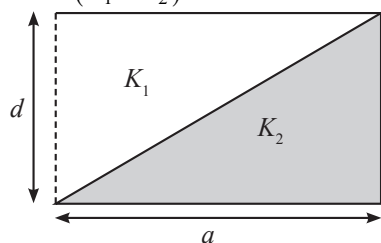
We know that, $C = \frac{\epsilon_0 A}{d}$

$$dC = \frac{\epsilon_0 a dx}{d + \alpha x} \Rightarrow \int dC = \int \frac{\epsilon_0 a dx}{d + \alpha x}$$

$$\Rightarrow C = \frac{\epsilon_0 a}{\alpha} [\ln(d + \alpha x)]_0^a$$

$$\Rightarrow C = \frac{\epsilon_0 a}{\alpha} \ln\left(1 + \frac{\alpha a}{d}\right) \approx \frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$$

$$17. (a) C_{eq} = \frac{\epsilon_0 K_1 K_2 a^2 \ln \frac{K_1}{K_2}}{(K_1 - K_2) d}$$

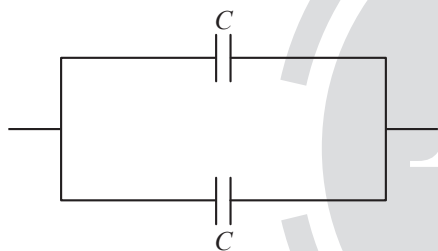


$$\Rightarrow C_{eq} = \frac{\epsilon_0 K a^2 \ln \frac{1}{K}}{d(1-K)}$$

As $K_1 = 1, K_2 = K$

$$\Rightarrow C_{eq} = \frac{\epsilon_0 K a^2}{d(K-1)} \ln K$$

18. (c) Reduced circuit is,

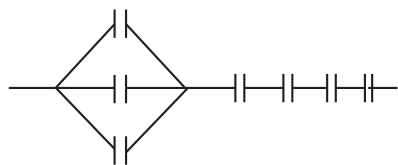


Both capacitors are in parallel.

$$\therefore C_{total} = C + C = 2C$$

$$19. (b) C_{eq} = \frac{6}{13} \mu F$$

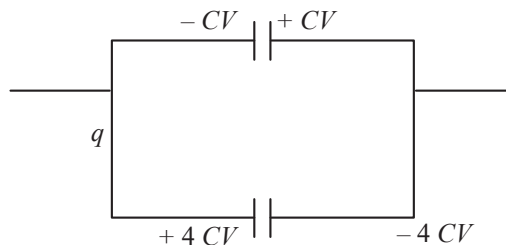
Therefore three capacitors must be in parallel to get 6 in



$$\frac{1}{C_{eq}} = \frac{1}{3C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C}$$

$$C_{eq} = \frac{3C}{13} = \frac{6}{13} \mu F$$

20. (a) Here, we have given two capacitor of capacitance C and $2C$



$$V_{common} = \frac{4CV - CV}{3C} = V$$

$$C_{eq} = C + 2C = 3C$$

$$\therefore V_f = \frac{1}{2} \times (3C) \times V^2 = \frac{3}{2} CV^2$$

21. (a) Equivalent capacitance,

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{10} + \frac{1}{15} + \frac{1}{20}$$

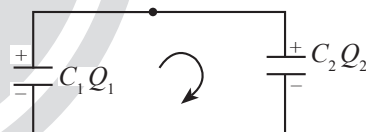
$$\frac{1}{C_{eq}} = \frac{6+4+3}{60} = \frac{13}{60}$$

$$C_{eq} = \frac{60}{13} \mu F$$

Total charge in the circuit

$$Q = C_{eq} V = \frac{60}{13} \times 13 \mu C = 60 \mu C$$

22. (a) After switch S is closed



$$Q_1 + Q_2 = C_1 V \quad \dots(i)$$

using kirchhoffs volatage low

$$\frac{Q_1}{C_1} - \frac{Q_2}{C_2} = 0$$

$$Q_1 = Q_2 \frac{C_1}{C_2} \quad \dots(ii)$$

from (i) and (ii)

$$Q_2 \frac{C_1}{C_2} + Q_2 = C_1 V \Rightarrow Q_2 = \left(\frac{C_1 C_2}{C_1 + C_2} \right) V$$

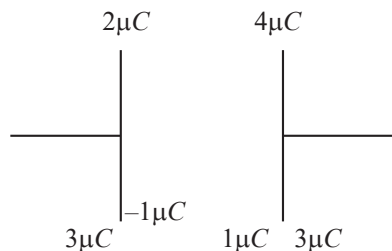
23. (a) $\Delta H = \Delta U$

$$= \frac{1}{2} \times \frac{C \times 3C}{C + 3C} (V^2)$$

$$= \frac{1}{2} \times \frac{3}{4} CV^2 = \frac{3}{8} CV^2$$

$$\Delta H = \frac{3Q^2}{8C}$$

24. (b)



$$\text{Charge at outer plates} = \frac{2+4}{2} = 3\mu C$$

$\therefore$ Charges at inner plates are $1\mu C$ and $-1\mu C$.

$$\text{Potential difference} = \frac{q}{C} = \frac{1 \times 10^{-6} C}{1 \times 10^{-6} F} = 1 V$$

25. [3] For $x = \frac{d}{3}$

$$C_1 = \frac{\epsilon_0 A}{\left(\frac{d/3}{k} + \frac{2d}{3}\right)} = \frac{\epsilon_0 A}{\frac{d}{12} + \frac{2d}{3}}$$

$$= \frac{\epsilon_0 A}{d} \times \left(\frac{12}{9}\right)$$

$$C_1 = \frac{4}{3} \frac{\epsilon_0 A}{d} = 2\mu F \Rightarrow \frac{\epsilon_0 A}{d} = \frac{6\mu F}{4}$$

$$\text{for } x = \frac{2d}{3}$$

$$C_2 = \frac{E_0 A}{\left(\frac{2d/3}{k} + \frac{d}{3}\right)} = \frac{E_0 A}{d} \times 2$$

$$\Rightarrow \frac{6}{4} \times 2 = 3\mu F$$

26. [23] Capacitance of parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d}$$

Where, d = distance between the plates of capacitor.

$$C_1 = \frac{\epsilon_0 A}{5b}$$

$$C_2 = \frac{\epsilon_0 A}{3b}$$

$$C_3 = \frac{\epsilon_0 A}{b}$$

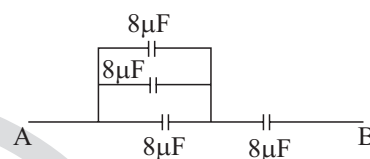
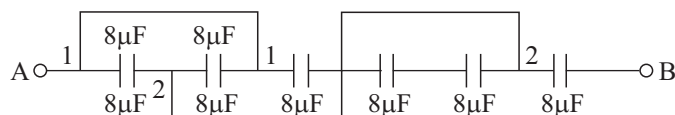
Since, capacitors are connected in parallel therefore,

$$C_{eq} = C_1 + C_2 + C_3$$

$$= \frac{\epsilon_0 A}{b} \left[\frac{1}{5} + \frac{1}{3} + 1 \right]$$

$$C_{eq} = \frac{23}{15} \frac{\epsilon_0 A}{b}$$

27. [6]



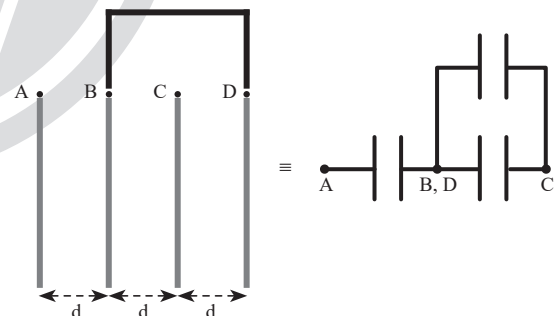
$$C_{eq} = \frac{24 \times 8}{32} = 6\mu F$$

28. [3] The given capacitor can be considered as two capacitor (one with dielectric and another without dielectric) connected in parallel.

$$\text{So, } \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2k\epsilon_0 A} + \frac{d}{2\epsilon_0 A}$$

$$\Rightarrow C_{eq} = \frac{2\epsilon_0 A k}{a(k+1)} = \frac{2 \times 2 \times 3.2}{4.2 \times 1} \epsilon_0 = 3.04 \epsilon_0$$

29. [2]



$$C_{AC} = C_{eg} = \frac{C \times 2C}{e + 2C}$$

$$= \frac{2}{3} C$$

$$C_{AC} = \frac{2\epsilon_0 A}{3d} = \frac{2\epsilon_0 \times 2 \times 10^{-2} \times \frac{3}{2} \times 10^{-2}}{3d} = \frac{2\epsilon_0}{d} \times 10^{-4} F$$

30. [16] Let charge on C_2 be $q\mu C$.

$$\text{Therefore, } \frac{q}{8} = \frac{2 \times 10 - q}{2} \Rightarrow q = 16$$